

Computer Networks - Lab

Final Exam

(CL3001)

Total Time (Hrs): 2.5 hrs

Date: 19th May 2026

Total Marks: 40

Lab Instructor

Total Questions: 3

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Roll No

Section

Student Signature

Read the instructions below carefully:

- Understanding the question statement is also part of the exam, so do not ask any questions. In case of any ambiguity, make suitable assumptions.
- You have to complete the exam in **2 hour 30 minutes** including the submissions; no late submissions will be acceptable whatsoever.
- Plagiarism and use of the internet is highly discouraged and can result in a disciplinary action.
- **Submission of output screenshots is mandatory. For questions related to the packet tracer please submit all screenshots of configurations, ping commands and each working/implementation step along with the .pkt file. In case you fail to comply with this you will be awarded 0 marks. Also please clearly label all IP addresses, AS names etc.**
- In case of missing or corrupted submission, **ZERO** marks will be awarded.
- Submission of exam in the correct folder is your responsibility, if you submit in the wrong folder or submit the wrong file, only you will be responsible and no excuses will be accommodated.
- Submission path is as follows: **//exam/final exam/computer networks lab/section/submissions**
- Zip your submission. Submit it to your respective folder using a drag and drop method. The new system disallows multiple submissions. After submitting your exam you can Enlist all the contents of the submission folder (you won't be able to view the contents), in case you plan to change your submission, delete your folder before submitting the updated version.
- You may crosscheck your submission by dragging and dropping your folder again, a message showing "Replace folder" would indicate a successful submission.

Question NO. 01: SOCKET PROGRAMMING TASK

[10 Marks]

A university library wants to automate its book issuing process using a network-based system. Multiple students should be able to connect to the server simultaneously and search for books by title. The server maintains a file named "Books_Collection.txt". This file contains records of books. Example format of "Books_Collections.txt":

OS
CN
DB
AI
ML

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Server Side Requirements

1. Create a TCP server that:
 - Listens for incoming client connections.
 - Supports multiple clients concurrently using multithreading.
 - Creates a separate thread for each connected client.
2. The server must:
 - Read book data from Books_Collection.txt.
 - Accept a book title sent by a client.
 - Search for the requested book.
3. If the book exists and is available:
 - Send a confirmation message to the client with book name and due date.

Example Success Response

Book Issued Successfully

Book: OS

Due Date: 29-May-2026

4. If the book exists but is already issued:
 - Send an appropriate error message.

Example

Error: Book is currently issued to another student.

5. If the book does not exist:
 - Send an appropriate error message.

Example

Error: Requested book is not available in library database.

6. The server should properly:
 - Handle multiple requests simultaneously.
 - Close client connections gracefully.
7. The server should remain active continuously until manually terminated.

Client Side Requirements

1. Create a TCP client application that:

- Connects to the server.
 - Allows a student to enter a book title.
 - Sends the title to the server.
 - Displays the server response.
2. The client should terminate gracefully after receiving the response.

Sample Execution

Client 1 in terminal 1

Enter Book Title: OS

Server Response displayed on client terminal 1

Book Issued Successfully

Book: OS

Due Date: 29-May-2026

Client 2 in terminal 2

Enter Book Title: OS

Server Response displayed on client terminal 2

Error: Book is currently issued to another student.

Client 3 in terminal 3

Enter Book Title: DS

Server Response displayed on client terminal 3

Error: Requested book is not available in library database.

You must submit:

1. Server.py file
2. Client.py file

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3. Output screenshots of server and clients (1 server terminal screenshot and 3 client terminals screenshots of all 3 scenarios mentioned above)

Question NO. 02: CISCO PACKET TRACER

[10 Marks]

You are required to design and implement a wired network infrastructure for a National Bank Branch Office using Cisco Packet Tracer.

The bank consists of the following departments:

- Customer Service
- Accounts Department
- Loan Department
- Human Resources
- IT Department

Each department must be placed on a separate subnet.

Network Requirements

The bank has a total of 100 computers distributed as follows:

Department	Number of Computers
Customer Service	20
Accounts Department	20
Loan Department	20
Human Resources	20
IT Department	20

You may add up to **two dummy PCs** to demonstrate proper subnet utilization.

IP Addressing Constraints

The bank is assigned a Class C network with the first three octets fixed as:

192.168.50.X

You are allowed to modify the last octet to create:

- Required number of subnets
- Required number of hosts per subnet

Apply optimal subnetting techniques.

Design & Implementation Constraints

- Use only wired connections:
 - Straight-through cables
 - Crossover cables where appropriate
- Wireless LAN is NOT allowed.

Use only the following networking devices:

- Cisco 2911 Routers
- Cisco 2950-24 Switches

A single router can connect to a maximum of two networks
(except router-to-router connections).

Configuration Requirements

1. Network Topology

Design a neat and organized network topology for the bank.

2. Static IP Addressing

Assign static IP addresses to:

- All PCs
- Router interfaces

3. Departmental Segmentation

- Each department must be placed on a separate subnet.
- Multiple subnets may be used within a department if needed.

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4. Device Optimization

Use the minimum number of routers and switches while satisfying all given constraints.

5. Documentation in Packet Tracer

Add proper labels/comments showing:

- Department names
- Subnet addresses
- Valid IP ranges
- Router interface IP addresses

Verification Requirements

Use ping commands to verify:

Intra-department Connectivity

- PCs within the same department should communicate successfully.

Inter-department Connectivity

- PCs from different departments should communicate through routers.

You must submit:

1. Cisco Packet Tracer (.pkt) file
2. Screenshot of complete network topology
3. Screenshot showing successful intra-department ping
4. Screenshot showing successful inter-department ping

Question NO. 03: CISCO PACKET TRACER

[20 Marks]

You are required to design and implement a network using Cisco Packet Tracer for a small organization. The network must support dynamic routing using OSPF and application-layer email communication using an SMTP mail server. The organization has three departments connected through three routers. Each department is placed in a separate subnet.

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Network Requirements

Devices Required

- 3 Routers (R1, R2, R3)
- 3 Switches
- 2 PCs in each subnet (PC1 and PC2 in subnet A, PC3 and PC4 in subnet B, PC5 and PC6 in subnet C)
- 1 SMTP Mail Server (located in Subnet C)

Subnet Structure

The network is divided into three subnets:

Subnet	Description	Devices
Subnet A	Client Network	PCs (Sender)
Subnet B	Intermediate Network	PCs
Subnet C	Server Network	SMTP Mail Server + Receiver PCs

Network Design & Subnetting

Using a given IP block (e.g., 192.168.1.0/24 or 10.0.0.0/16):

1. Perform proper subnetting to create three separate subnets.
2. Assign IP addresses to:
 - All PCs
 - Server
 - Router interfaces
3. Ensure each subnet has:
 - Network ID
 - Usable IP range
 - Default gateway

Topology Design

Design the network using the following structure:

- Router 1 connects Subnet A
- Router 2 connects Subnet B (acts as intermediate router)

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- Router 3 connects Subnet C (server network)

Physical Layout:

- Each subnet has:
 - 1 Switch
 - End devices (PCs or server)
- Routers are connected in a chain:
 - R1 ↔ R2 ↔ R3

OSPF Configuration

Configure **OSPF (Open Shortest Path First)** on all three routers.

Requirements:

- Enable OSPF process on all routers (Attach screenshot of OSPF setting configuration on any router)
- Advertise all connected networks

Verification:

- Check routing tables using:
 - show ip route
- Check OSPF neighbors using:
 - show ip ospf neighbor
- Verify connectivity using:
 - Ping between PCs in different subnets (Ping from subnet 1 PC to subnet 2 and 3 PCs)

SMTP Mail Server Configuration

Mail Server Setup (Subnet C)

1. Configure the SMTP Mail Server
2. Create email accounts

Client Configuration

1. On PC in Subnet A, configure email client
2. On PC in Subnet C, configure recipient email account

Email Communication Demonstration

Demonstrate application-layer communication:

1. Send an email from **Subnet A client** to user in Subnet C (Attach screenshot)
2. Email must pass through:
 - o Router 1 → Router 2 → Router 3 (via OSPF routing)
3. Verify email successfully received (Attach screenshot)
4. Reply to email and verify successful receiving of reply in client PC (Attach screenshot of reply received)

Verification Requirements

Submit screenshots of:

- Network Topology and OSPF Setting configurations
- OSPF Verification (show ip route, show ip ospf neighbor)
- Connectivity Tests (Ping from Subnet A PC to Subnet B and C PC)
- SMTP Verification (Email sent, email received, reply received)

Submission Requirements

You must submit:

- Cisco Packet Tracer file (.pkt)
- Network topology screenshot
- Ping screenshots (From subnet A PC to subnet B and C PCs)
- OSPF configuration screenshots (both setting and verification as mentioned above)
- Email send, receive and reply receive screenshots

Properly rename each screenshot or if you are submitting a word file then give proper headings.

```
en
config t
hostname R1

int f0/0
ip add 192.168.12.1 255.255.255.0
no sh

int f0/1
ip add 192.168.14.1 255.255.255.0
no sh

int f1/0
ip add 192.168.1.1 255.255.255.0
no sh

router ospf 1

int f0/0
ip ospf 1 area 0
int f0/1
ip ospf 1 area 0
int f1/0
ip ospf 1 area 0
```

OR

```
Router(config)#router ospf 30
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 200.100.50.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```